

Installation and Commissioning Instructions

MTU Integrated Automation
FAL 6 Components
(Field Automation Level)

E532306/02E



Power. Passion. Partnership.

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1 Module design

1.1 FAL MTU Integrated Automation terminals – Overview of functions

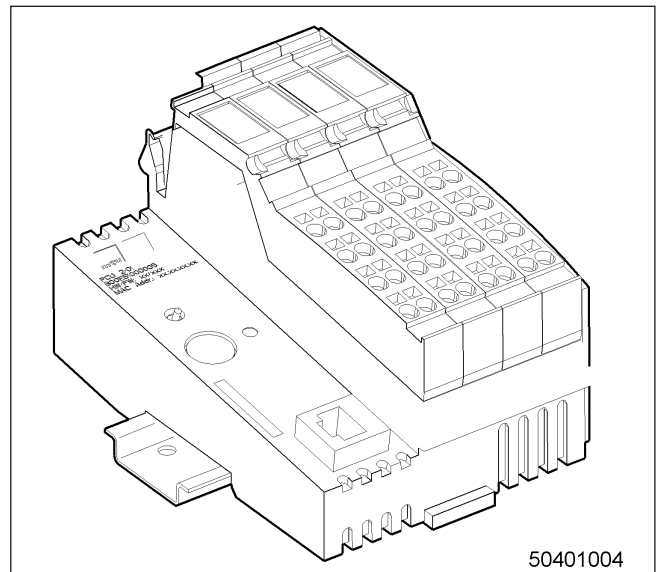
Field bus level Interbus and Profinet

Devices offering various functions are available for the Interbus and Profinet field bus types:

Controller	Programmable Control Device
Interbus	PCD 2, PCD 3, PCD, 4
Profinet	PCD 3, PCD 4
Bus coupler	Bus Interface Device
Interbus	BID 1, BID 2
Profinet	BID 3
Serial interface terminals	Serial Interface Device
Interbus	SID 1, SID 2, SID 3
Profinet	SID 1, SID 2, SID 3
I/O terminals	Interbus and Profinet
Binary	DID 1, DID 2 - Digital Input Device
	DOD 1, DOD 2, DOD 3, DOD 4 - Digital Output Device
Analog	AID 1, AID 2, AID 3 - Analog Input Device
	TID 1, TID, 2 TID 3 - Temperature Input Device
	AOD 1, AOD 3 - Analog Output Device
Function terminals	Interbus and Profinet
Analog	FID 1 - Frequency Input Device
	PWD 1 - Pulse Width Device
Infeed terminals	Interbus and Profinet
Supply	PID 1, PID 2- Power Input Device

Controller (control terminals)

Various controllers (PCDs) are available for use as control terminals with decentralized intelligence.



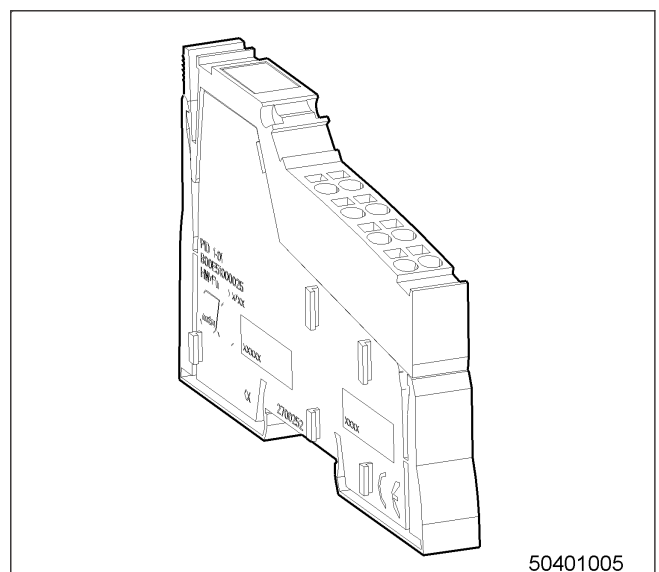
PCD controllers belong to the group of Remote Field Controllers (RFC). The MTU substation becomes a decentralized controller when equipped with one of these controllers. Each controller integrates a control CPU which can be programmed in IEC61131 in the automation system. It facilitates direct processing of signals from the decentralized input/output points. Independent sub-networks which process automation tasks on their own can be created with a PCD controller.

Infeed terminals

The requisite voltages are supplied to the potential routers in the station via an infeed terminal. Several infeed terminals can be used in one station. This allows electrical isolation between the various power circuits and the creation of sectors having differing voltages inside the station.

The main and segment voltages are supplied to terminal PID 1.

An additional 24 V supply voltage is supplied to the PID 2 terminals from which the logic voltage U_L is generated.



Potential router

The infeed terminal interrupts all potential routers for the voltages which are newly supplied and re-establishes them.

Load-bearing capacity of the router contacts

The maximum current-bearing capacity of the lateral router contacts is specified in the section entitled “Current and voltage distribution”.

Electrical isolation

The infeed terminal is used to create floating peripheral sectors within a substation.

Functional grounding (24 V DC)

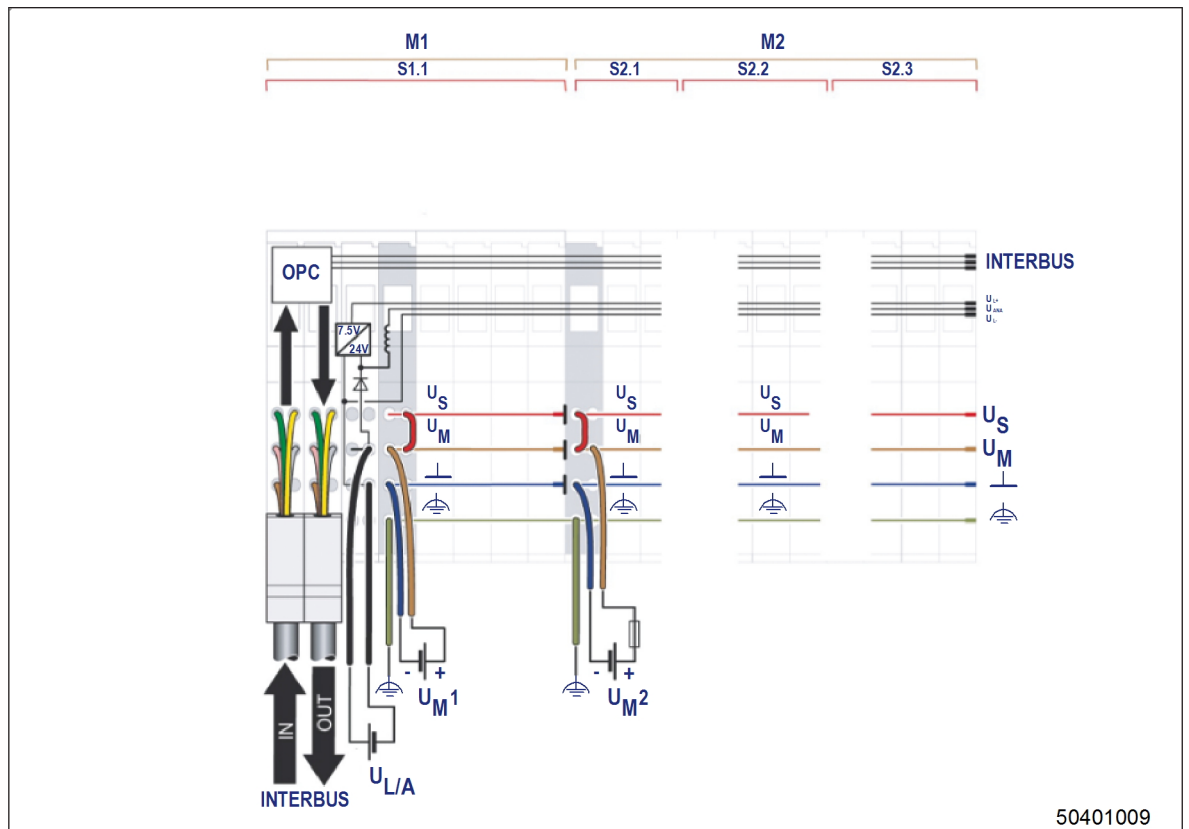
Connection to functional ground is established by the FE springs on the bottom of the infeed terminals on the 24 V level when the terminal is snapped onto the grounded mounting rail. These springs are connected to potential router FE and with the terminal points for an FE connection.

If the preceding terminal is a terminal on the 24 V level, the infeed terminal is simultaneously connected to the potential router FE of the substation when snapped onto the preceding terminal.

Reference documentation

Detailed information about the function, characteristics and wiring of the various terminals is provided in the documents pertaining to the specific terminals.

Infeed and segmentation



M1 Main power circuit 1

M2 Main power circuit 2

OPC OPC protocol chip (bus logic including voltage conditioning)

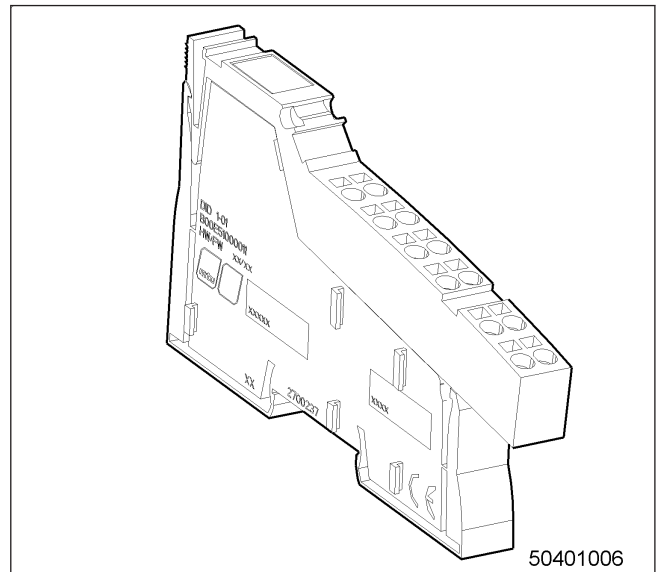
The infeed terminals create different segment circuits within a 24 V sector of an MTU substation (segment terminals are not available in the present scope of delivery).

Input/output terminals

Terminals with a variety of functions are available for the small signal level. The following terminals are examples. The article designation corresponding to the respective function is written in parentheses:

- Digital input and output terminals (DID, DOD)
- Analog input and output terminals (AID, AOD)
- Relay terminals (DOD)
- Temperature acquisition terminals (TID)
- Function terminals (FID)

These terminals are available in various sizes. This allows modular design to tailor substations for specific applications.



Fuse

Each MTU substation (comprising a series of MTU Integrated Automation terminals) must be protected separately. The rating of the upstream fuse must be selected such that it does not exceed the maximum load current. Refer to the terminal-specific documents for details of the max. admissible load current of the I/O terminal.

Load-bearing capacity of the router contacts

The maximum current-bearing capacity of the lateral router contacts is specified in the section entitled “Current and voltage distribution”.

Grounding (FE or PE)

Functional grounding (24 V DC sector) is established by the relevant potential router when snapped onto the preceding terminal.

Shielding

Special shield connectors are available to connect shielded lines.

Parameterization

Some terminals can be parameterized by process data or PCP. Refer to the terminal-specific documents for details.

Data formats

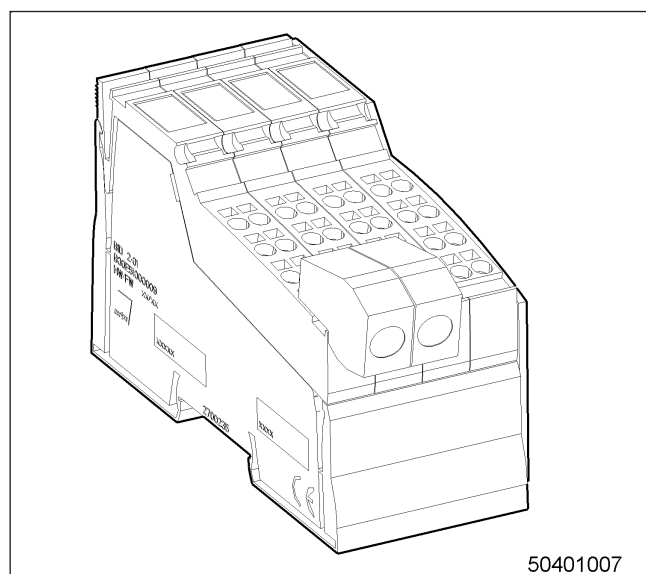
Depending on the terminal used and its configuration, measured values and the corresponding output values of the analog and temperature acquisition terminals can be represented in various data formats. The formats are described in the relevant terminal-specific documents.

Diagnosis

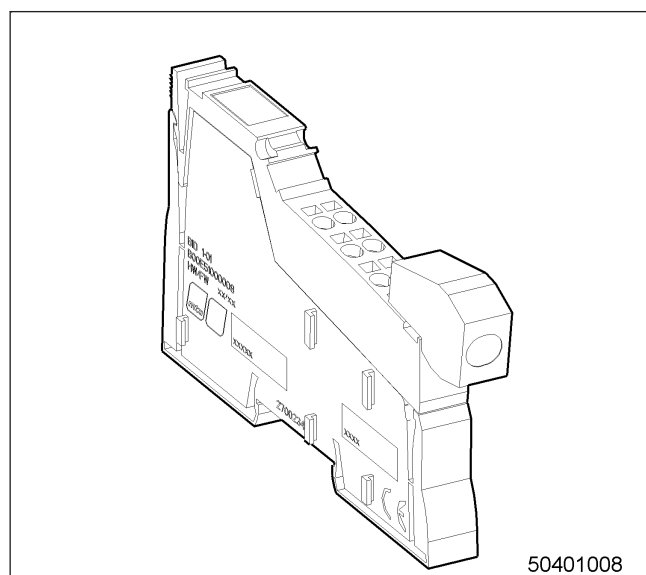
The scope of diagnostic functions depends on the terminal used and is specified in the relevant terminal-specific document.

Bus couplers and terminals with remote bus tap

Bus coupler for INTERBUS: BID 2



Terminal with remote bus tap: BID 1



Bus coupler

A bus coupler is used to connect a substation to the field bus. Bus couplers are available for various bus systems.

Functions

A bus coupler is used to amplify signals (repeater function), divide the system into segments and allow connection or disconnection of the outgoing remote bus and/or connected station during operation.

A bus coupler forms the head-end of an MTU substation. It couples the substation to the INTERBUS remote bus.

A bus coupler creates the logic voltage U_L for the connected stations, and the supply for the connected analog terminals U_{ANA} from the infeed supply voltage $U_{L/A}$.

Number of stations

The maximum number of stations which can be connected to a bus coupler is determined by the following parameters:

1. The maximum number of stations on the INTERBUS is limited to 512.
2. As many as 63 stations can be connected to a bus coupler. This number includes all stations downstream of the bus coupler, i.e. both the terminals and the devices which are connected via a branch terminal. The stations of any connected remote bus tap do not influence the number of stations in the substation.
3. The maximum current which the bus coupler is capable of supplying in the logic sector (U_L) is limited.
4. The current carrying capacity of the potential routers is limited. The limit values for the various potential routers are specified in the relevant sections.

The values specific to each particular terminal must be observed when configuring a substation. These are described in the relevant terminal-specific documents. The possible number of stations which can be connected depends on the specific design of the station. None of the parameters specified above may be violated.

End plate

The mechanical end plate is included in the bus coupler scope of delivery. The end plate terminates a substation and must be installed after the last terminal of the station. It has no electrical function. It protects the station from electrostatic discharge and the operator from contact with hazardous voltages.

Fuse

The bus coupler is equipped with elements for protection against reverse-polarity and transient overvoltage of the supplied voltages.

Functional grounding

Grounding is established by the FE springs on the bottom of the terminals when the terminal is snapped onto the grounded mounting rail. These springs are connected to potential router FE and with the terminal points for an FE connection.

Compulsory additional functional grounding

The bus coupler must be additionally connected to functional ground via the FE connection in order to ensure reliable functional grounding of the station if the FE springs become dirty or damaged. The terminal points for FE connection must be connected to a grounded PE terminal for this.

Load-bearing capacity

The maximum current-bearing capacity of the lateral router contacts is specified in the section entitled "Electrical potential and data routing".

Potential grounding

The various electrical ranges within a bus coupler are shown by examples in the figures below. The first figure illustrates electrical isolation in the bus coupler. Internal circuitry is not significant in this case. This is presented in the terminal-specific documents.